Huawei Access Node WFD-ACC-09 Whitefield POP Hardware/Environmental Fault Resolved by Huawei CLI

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8151

Description

The Huawei Access Node WFD-ACC-09 Whitefield POP Hardware/Environmental Fault failure domain is a result of the confluence of two critical factors. Firstly, the inherent susceptibility of the Huawei ASIC memory to parity errors is exacerbated by the environmental conditions at the Whitefield POP site. The site's high humidity and temperature fluctuations create an ideal breeding ground for hardware component degradation, leading to the manifestation of ASIC memory parity errors. Secondly, the role of the Access node in the network topology introduces an additional layer of complexity, as it is responsible for the aggregation and forwarding of traffic from multiple user networks. In the event of a line card crash or power supply redundancy loss, the node's ability to maintain network connectivity is severely compromised, resulting in the ALARM_8151 alarm being triggered. This failure domain is further compounded by the fact that the Huawei CLI dialect used for diagnostic and resolution commands is not as intuitive as other vendors, requiring a deeper understanding of the underlying network architecture and hardware mechanics.

Pre-requisite Checks

- Prerequisite: Verify site location and node ID
- Prerequisite: Check redundant link status and power supply redundancy
- Prerequisite: Dispatch remote hands to Whitefield POP for physical access

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display this, Expected Output: Huawei WFD-ACC-09 whitefield pop system information, Description: This command displays the system information of the Huawei Access Node WFD-ACC-09, including the node ID, site location, and hardware configuration.

Command: display fault, Expected Output: Fault information for WFD-ACC-09 whitefield pop, Description: This command displays the fault information for the Huawei Access Node WFD-ACC-09, including the
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alarm code, severity, and root cause of the fault.

Command: display env, Expected Output: Environmental information for WFD-ACC-09 whitefield pop, Description: This command displays the environmental information for the Huawei Access Node WFD-ACC-09, including temperature, humidity, and power supply status.
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Resolution Steps

Step 1: Step: Replace faulty line card, Command: replace linecard 0/0/0, Description: Replace the faulty line card with a new one to restore network connectivity.

Step 2: Step: Reboot node, Command: reboot node, Description: Reboot the node to ensure that all hardware components are functioning correctly.

Step 3: Step: Verify power supply redundancy, Command: display power, Description: Verify that the power supply redundancy is functioning correctly to prevent future faults.

Post-Check Verification

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Command: display this, Expected Output: Huawei WFD-ACC-09 whitefield pop system information, Description: Verify that the system information of the Huawei Access Node WFD-ACC-09 is correct after resolution.

Command: display fault, Expected Output: No fault information for WFD-ACC-09 whitefield pop, Description: Verify that the fault information for the Huawei Access Node WFD-ACC-09 is correct after resolution.
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Roll-back Procedures

- Step: Restore original configuration, Command: undo replace linecard 0/0/0, Description: Restore the original configuration to ensure that all hardware components are functioning correctly.

- Step: Reboot node, Command: reboot node, Description: Reboot the node to ensure that all hardware components are functioning correctly.